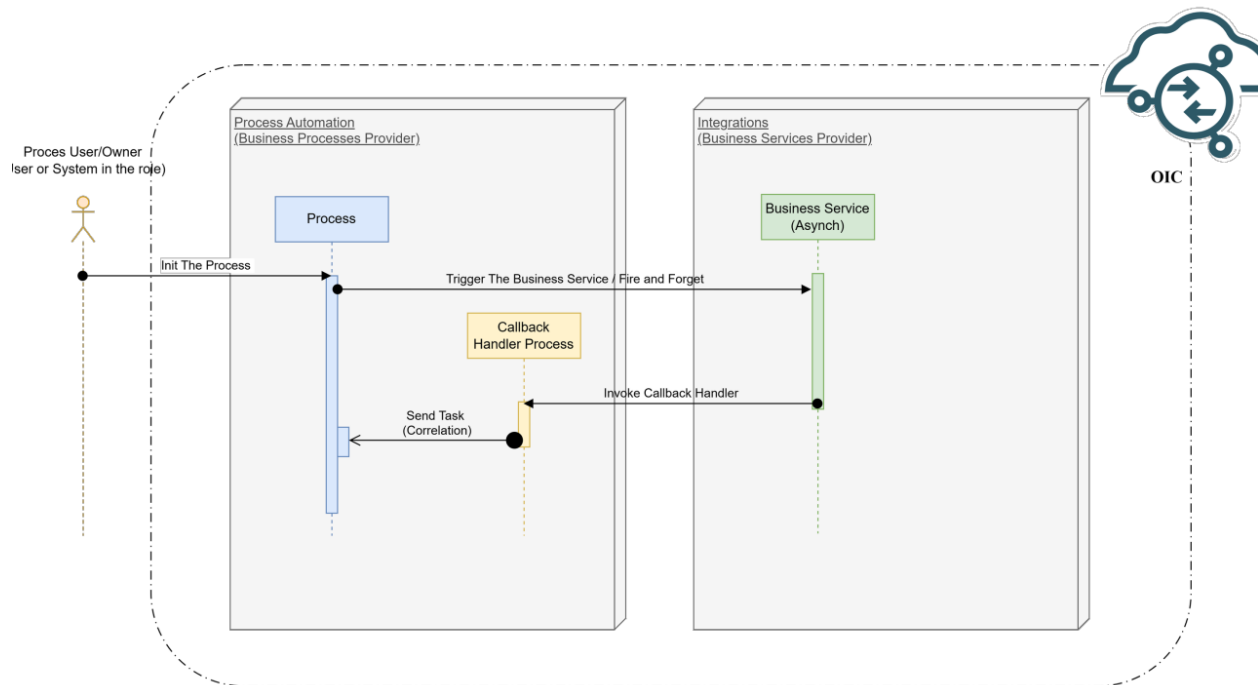




Synchronous → Asynchronous Business Orchestration



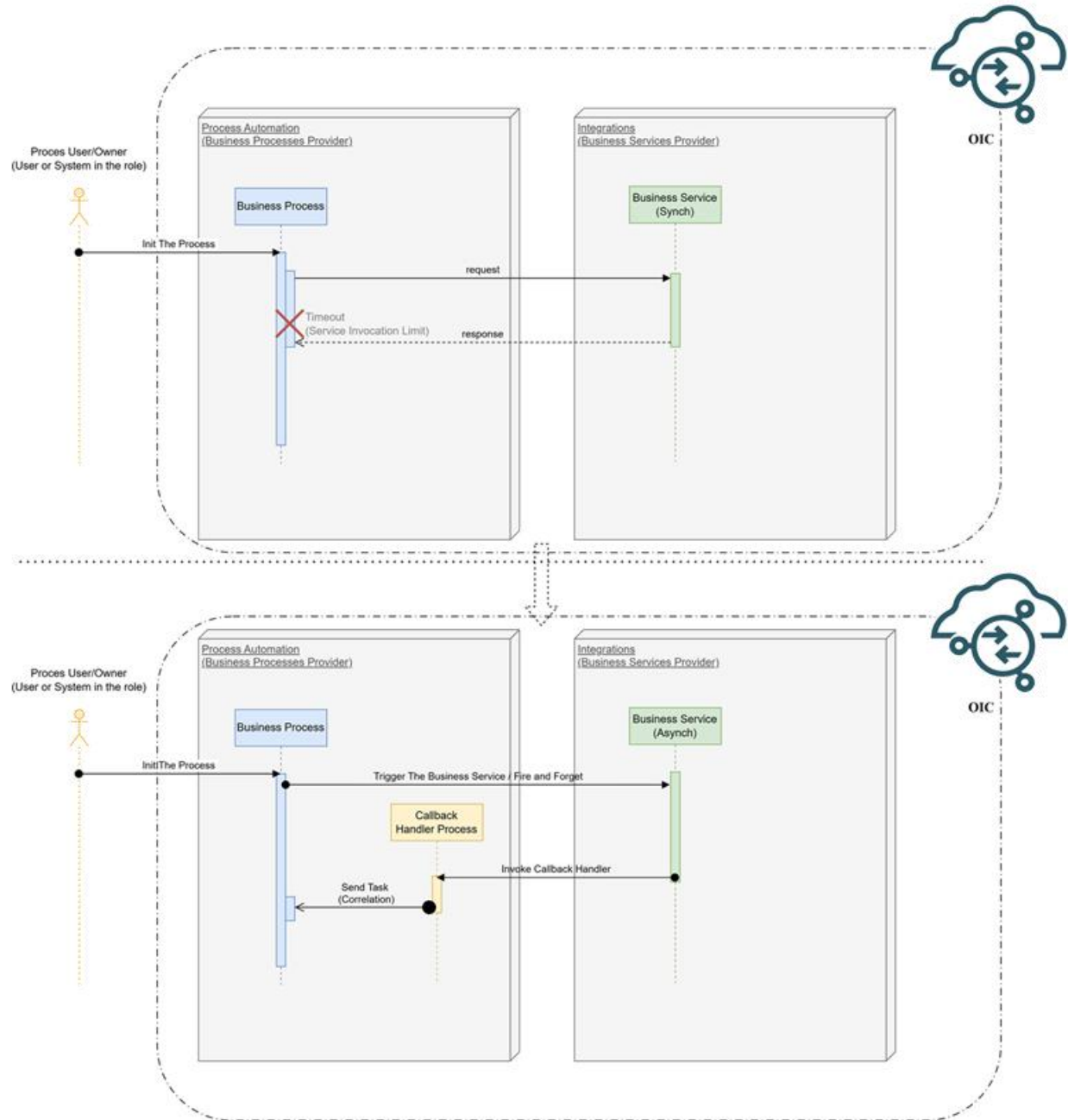
Peter Obert

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March 20, 2026

As business services mature over the time, their response times tend to grow. Synchronous orchestrations that once completed within acceptable limits may begin to time out, causing process failures.

This article walks you through the recipe for re-engineering those orchestrations into a robust asynchronous pattern using Oracle Integration Cloud (OIC) Process Automation and Integrations.



⚠ Problem

Long-running business services exceed synchronous timeout limits, causing faulted process instances in OIC Process Workspace.

✓ Solution

Refactor synchronous Integration flows and Process Automation orchestrations into an asynchronous callback pattern — achievable in minutes with OIC.

□ Limit

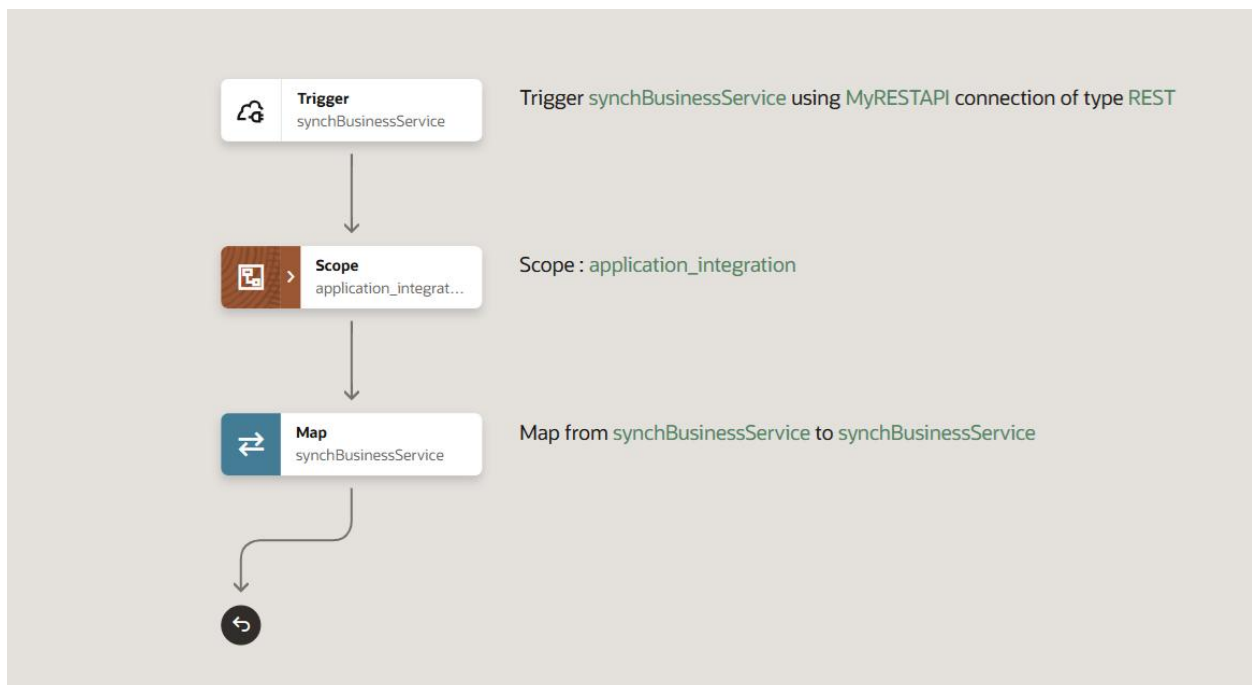
The OIC OPA HTTP Connector Timeout is currently 90 seconds. For services that respond in minutes, always design asynchronously. ([OPA Connector Timeout Limit](#))

Part 1 — Refactor the Integration (Make It Asynchronous)

Business services should be wrapped in Integration orchestrations to isolate technology-specific interactions and error handling from the business process itself. This design makes it safe to refactor the integration independently.

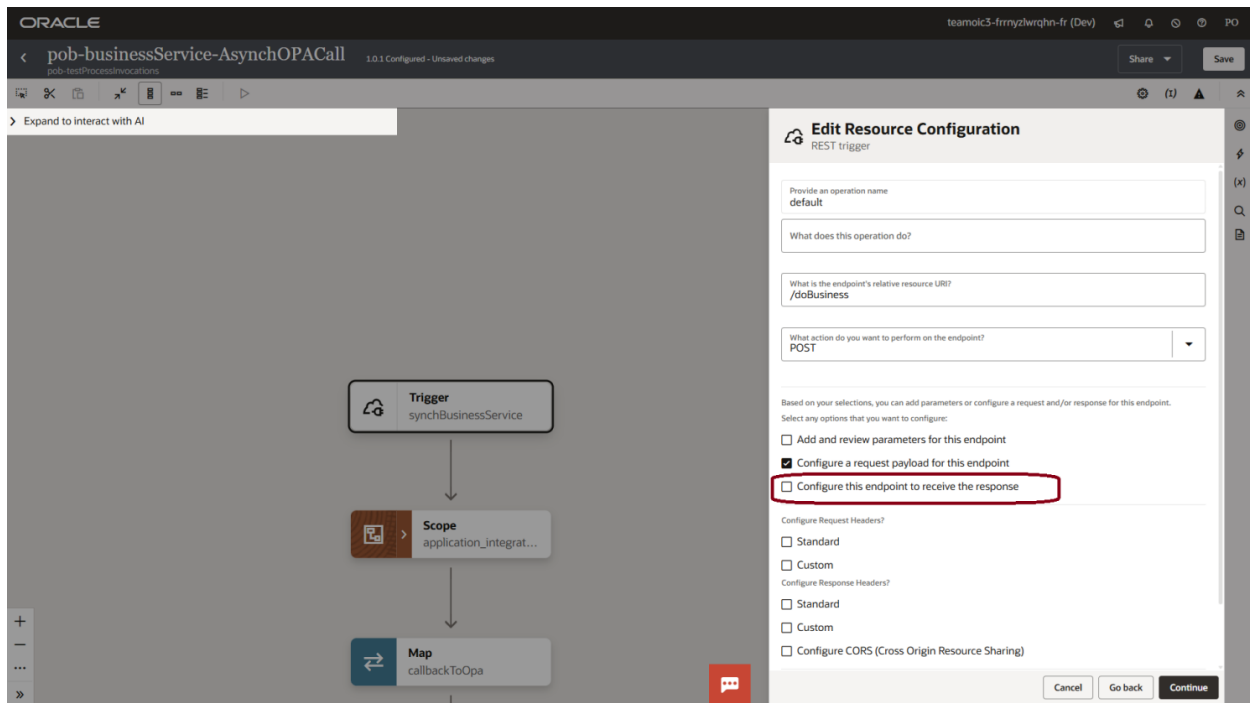
1 - Clone the existing synchronous Integration

Create a new version or clone of the integration that currently calls your long-running business service.



2 - Remove the response endpoint

In the cloned integration, open the trigger endpoint configuration and uncheck "Configure this endpoint to receive the response". This removes the Reply action and its associated mapping.

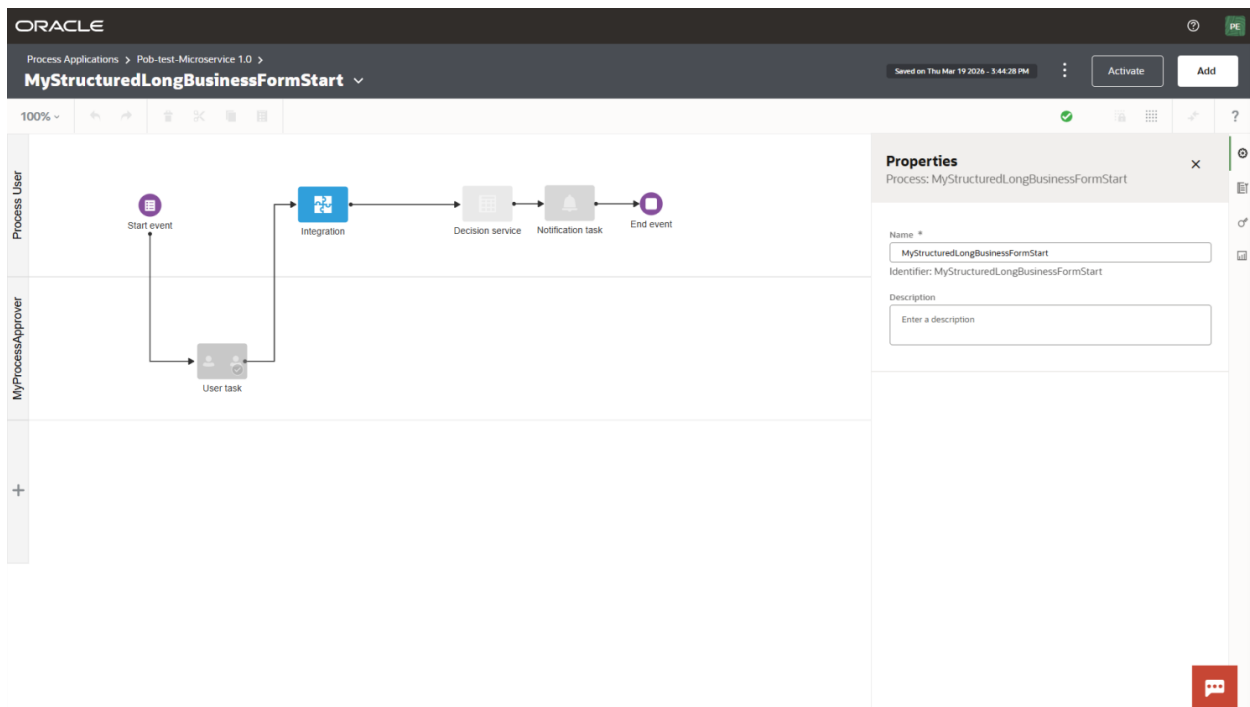


3 - Activate the new integration

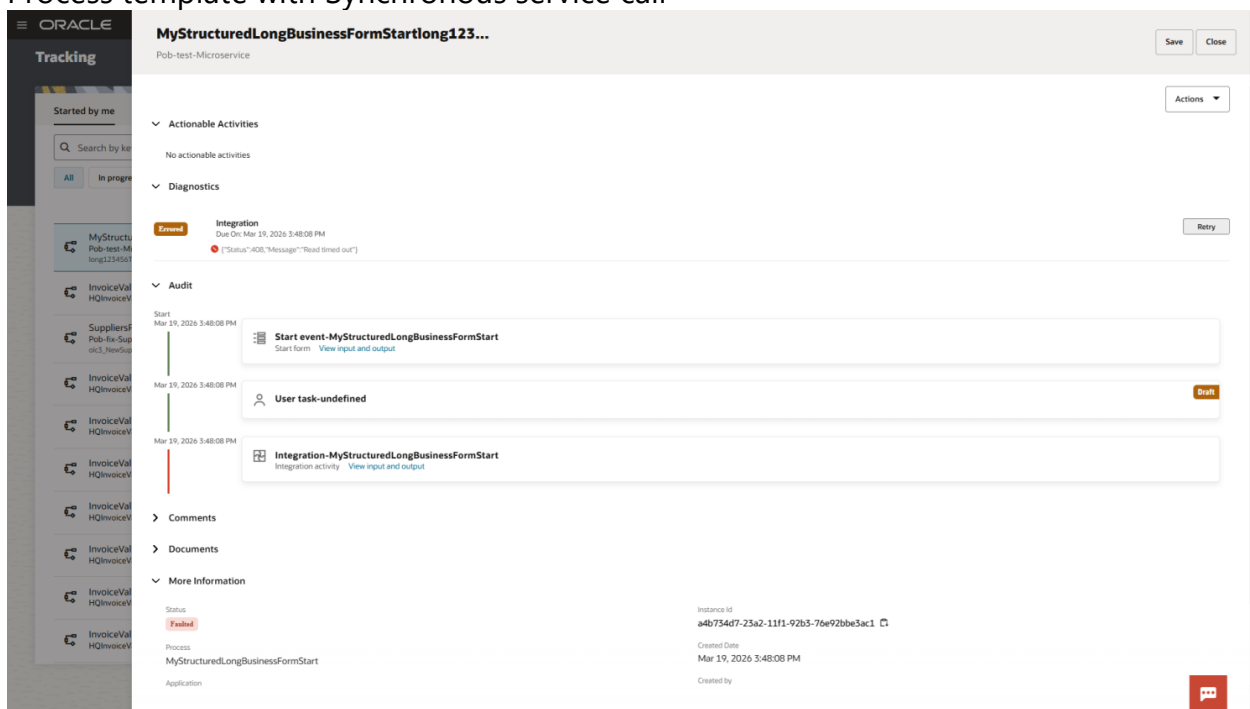
Save, validate, and activate the cloned asynchronous integration. Keep the original synchronous version active until the new process is fully validated.

Part 2 — Refactor the Existing Process (Call Integration Asynchronously)

The original process uses an Invoke Integration Service action that calls the synchronous integration. Faulted instances visible in Process Workspace are a symptom of timeout breaches.



Process template with Synchronous service call



Example of faulted synchronous call business service activity

The refactored process replaces the synchronous call with an async invocation followed by a correlated Receive Task.

1 - Create a connector for the async Integration

In OIC Process Automation, create a new connector pointing to the newly activated asynchronous integration.

Add component

BackCreate

Select integration context

☒ Project ☐ Not in project

pob-testProcessInvocations

▼

Select an integration. Only active integrations that expose REST endpoints appear in this list. [Help](#)

Q

Search for integrations

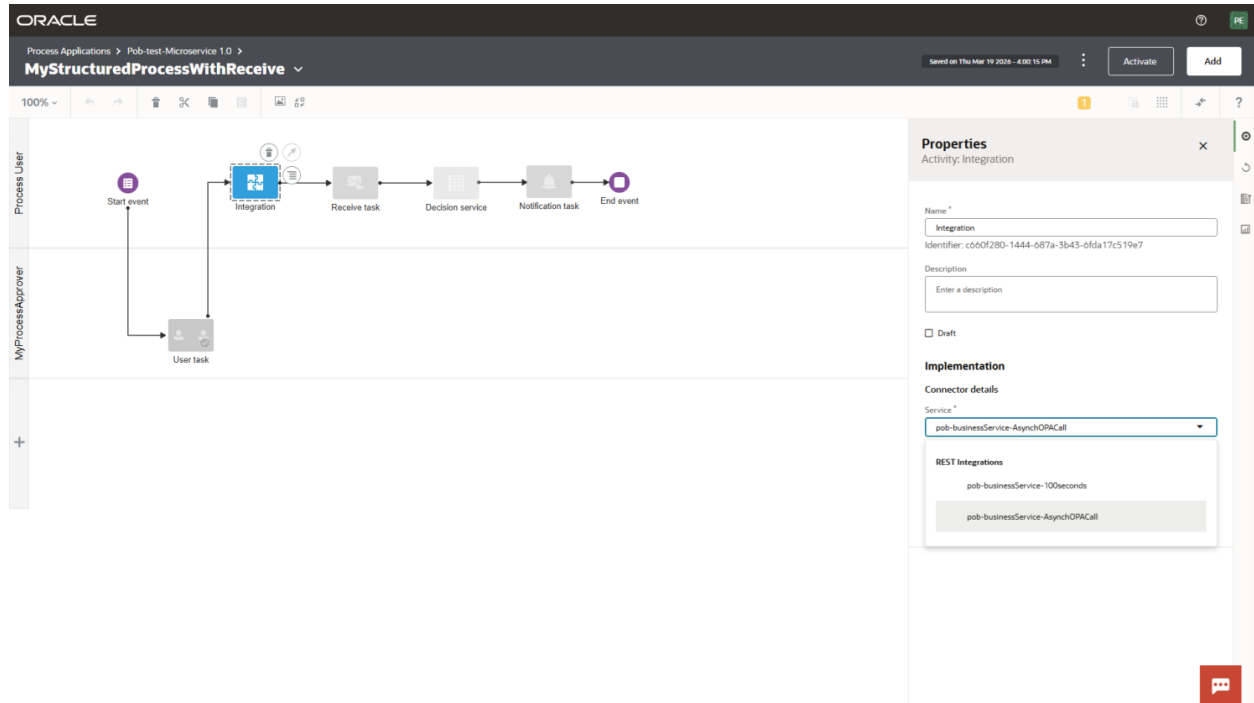
v01.00.0001	pob-businessService-AsynchOPACall OPA-rest-sync to async reengineering Synchronous long running service prototyper	Updated 59 minutes ago
v01.00.0001	pob-businessService-100seconds OPA-rest-sync to async reengineering Synchronous long running service prototyper	Updated 1 hour ago
v01.00.0000	myRestAPIInvokesNativeOPAServiceStringArray	Updated yesterday

2 - Clone the original Process

Create a clone of the business process. This preserves the original while you build and test the async variant.

3 - Replace the Invoke action with the async connector

In the cloned process, replace the original synchronous Invoke Integration Service action with the new async connector. Note: the invoke will no longer return any output.

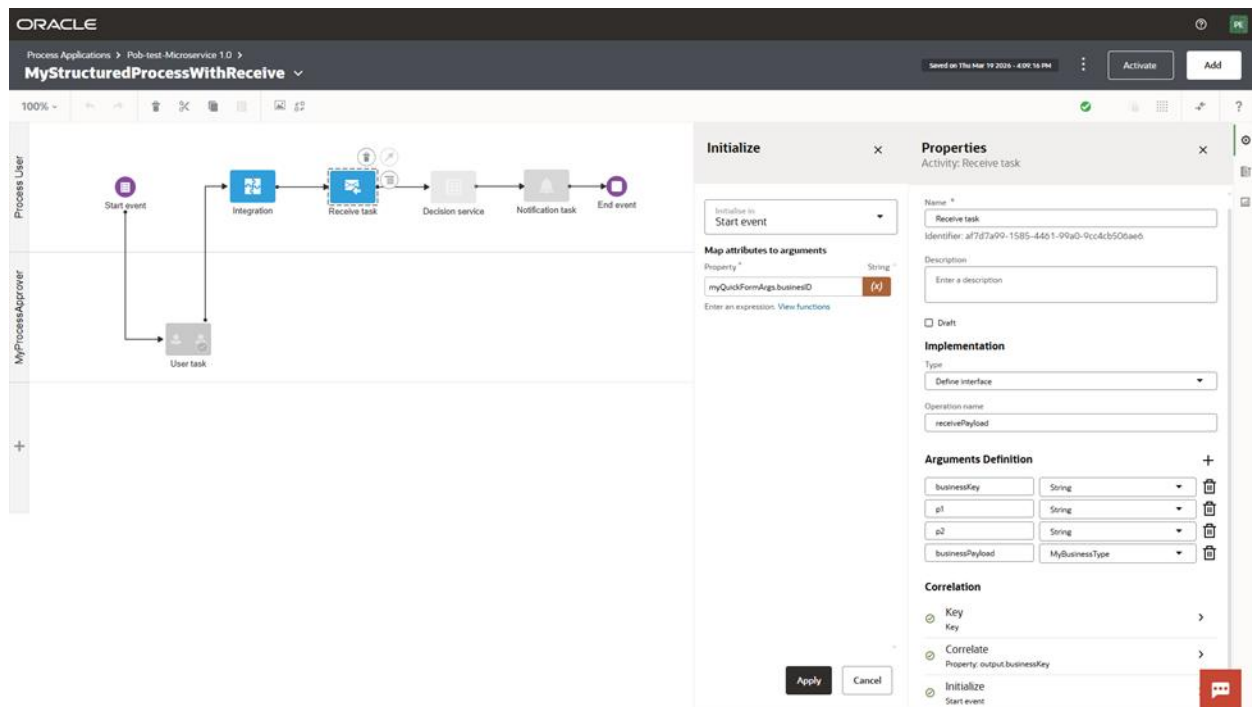


4 - Add a Receive Task after the invoke

Insert a Receive Task activity immediately after the async integration invoke. Configure it to receive the same business object that was previously returned as the integration output.

5 - Set correlation properties

On the Receive Task, define correlation properties so that incoming callback messages are matched to the correct running process instance. This is essential for routing the response to the right execution context.



Part 3 — Implement the Callback Handler (Process Automation)

The Callback Handler is a lightweight "microprocess" — a message-initiated Structured Process whose sole responsibility is to deliver the business object to the waiting Receive Task in the main process.

1 - Create a new Structured Process

Create a new message-initiated Structured Process. This will serve as the Callback Handler. Keep it minimal — it should contain only the input business object definition.

2 - Define the input Business Object

Add the business object as the input to the Callback Handler. This must match the object structure expected by the Receive Task in the refactored business process.

ORACLE

Process Applications > Pdb-test-Microservice 1.0 >

MyCallbackToProcess

Saved on Thu Mar 19 2020 - 4:11:57 PM

Activate Add

100%

Process User

Start event → Send task → End event

Properties

Activity: Start event

Name *

Start event

Identifier: 32206160-1ba0-11f1-a1ec-0d2819c910a3

Description

Enter a description

Who can run this activity?

Role members with at least use permission

Requires user authentication.

Arguments Definition

key	String
p1	String
p2	String
businessPayload	MyBusinessType

Instance

Define properties for instances of this process.

Title *

"MyCallbackToProcess" + process.businessKey

Enter an expression. View functions

Business key

Expression *

key

Enter an expression. View functions

3 - Send the Task to the waiting Process

Configure the Callback Handler to send the task to the Receive Task in the cloned business process. No reply is sent back to the business service — the handler's job ends here.

ORACLE

Process Applications > Pdb-test-Microservice 1.0 >

MyCallbackToProcess

Saved on Thu Mar 19 2020 - 4:15:21 PM

Activate Add

100%

Process User

Start event → Send task → End event

Properties

Activity: Send task

Name *

Send task

Identifier: c4d30e7f-c374-72b8-235e-73fb3a089eb3

Description

Enter a description

☐ Draft

Implementation

Type

Process call

Process

MyStructuredProcessWithReceive

Target mode

Receive task

Start event

Receive task

4 - Activate the Process Application with the Callback Handler

Save, validate, and activate the Callback Handler process before testing the end-to-end flow.

Part 4 - Business Service Callback to “Process Callback Handler Flow” (Integration)

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The screenshot displays the Oracle Process Applications interface for a process named "MyCallBackToProcess". The process flow is shown on the left, consisting of a "Start event" (purple circle), a "Send task" (blue square), and an "End event" (purple circle). The "Properties" panel on the right shows the configuration for the "Start event" activity.

Properties
Activity: Start event

Name: *
Start event

Identifier: 32206160-1ba0-11f1-a1ec-0d2819c910a3

Description
Enter a description

Who can run this activity?
Role members with at least use permission
Requires user authentication.

Arguments Definition

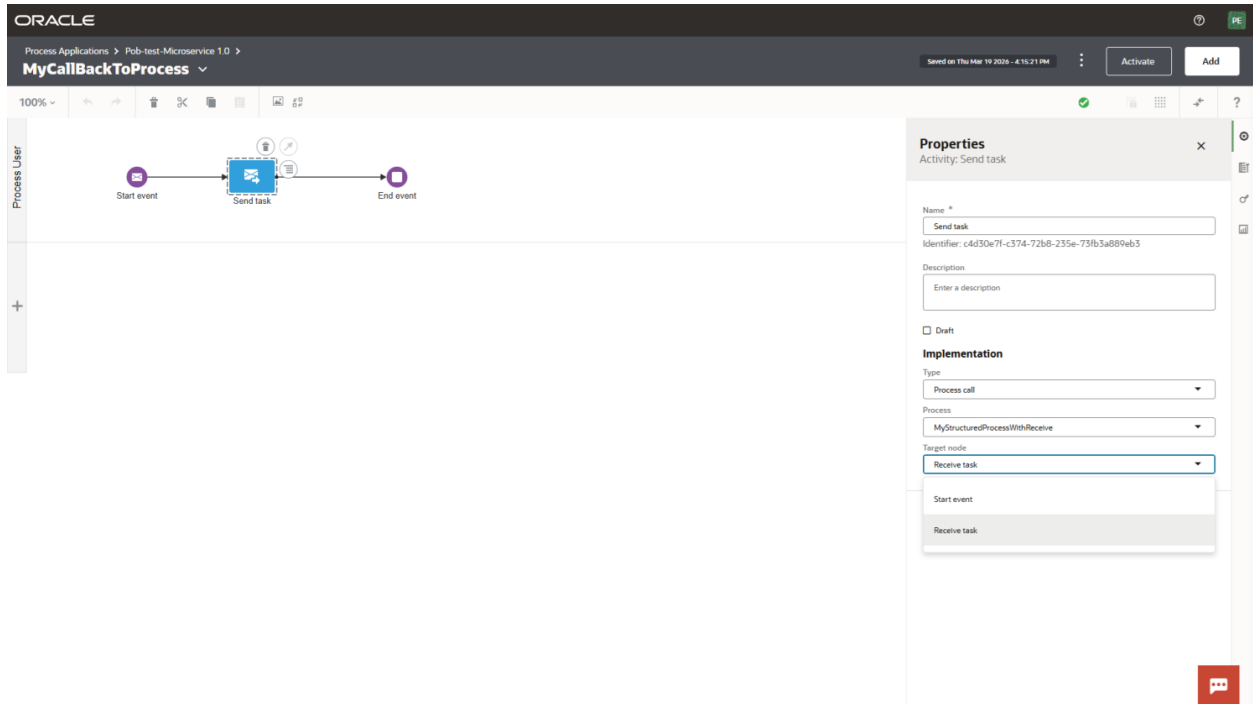
key	String
p1	String
p2	String
businessPayload	MyBusinessType

Instance
Define properties for instances of this process.
Title: *
"MyCallBackToProcess" + process.businessKey (x)

Business key
Expression: *
key
Enter an expression. View functions

3 - Send the Task to the waiting Process

Configure the Callback Handler to send the task to the Receive Task in the cloned business process. No reply is sent back to the business service — the handler's job ends here.



4 - Activate the Process Application with the Callback Handler

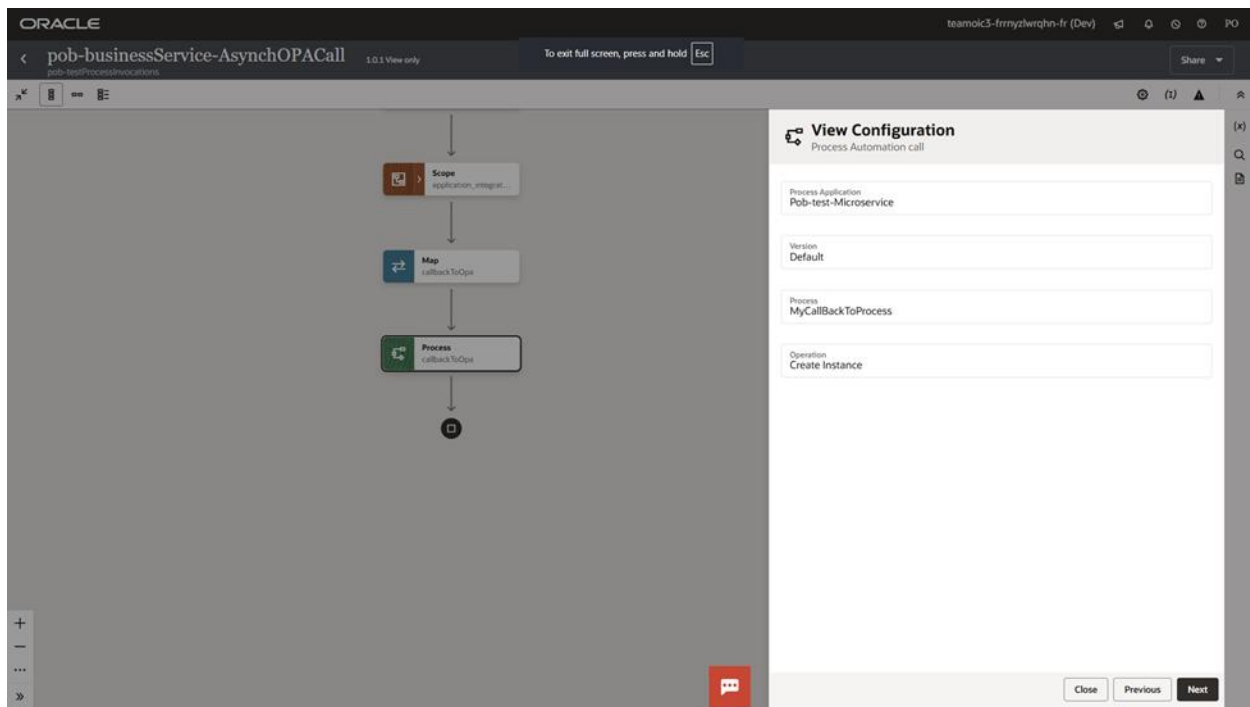
Save, validate, and activate the Callback Handler process before testing the end-to-end flow.

Part 4 - Business Service Callback to “Process Callback Handler Flow” (Integration)

Now we can finish the implementation of the newly created asynchronous Integration flow with invocation of the Process Callback Handler Flow.

1 - Add a Callback invocation

At the end of the async integration flow, add an invocation to the Process Callback Handler Flow (created in Part 3). Map the response payload exactly as the original mapping did — the data object structure remains the same.



Mapping implementation generated by the process invocation will be very the same as it was in the original mapping to populate "response" variable.

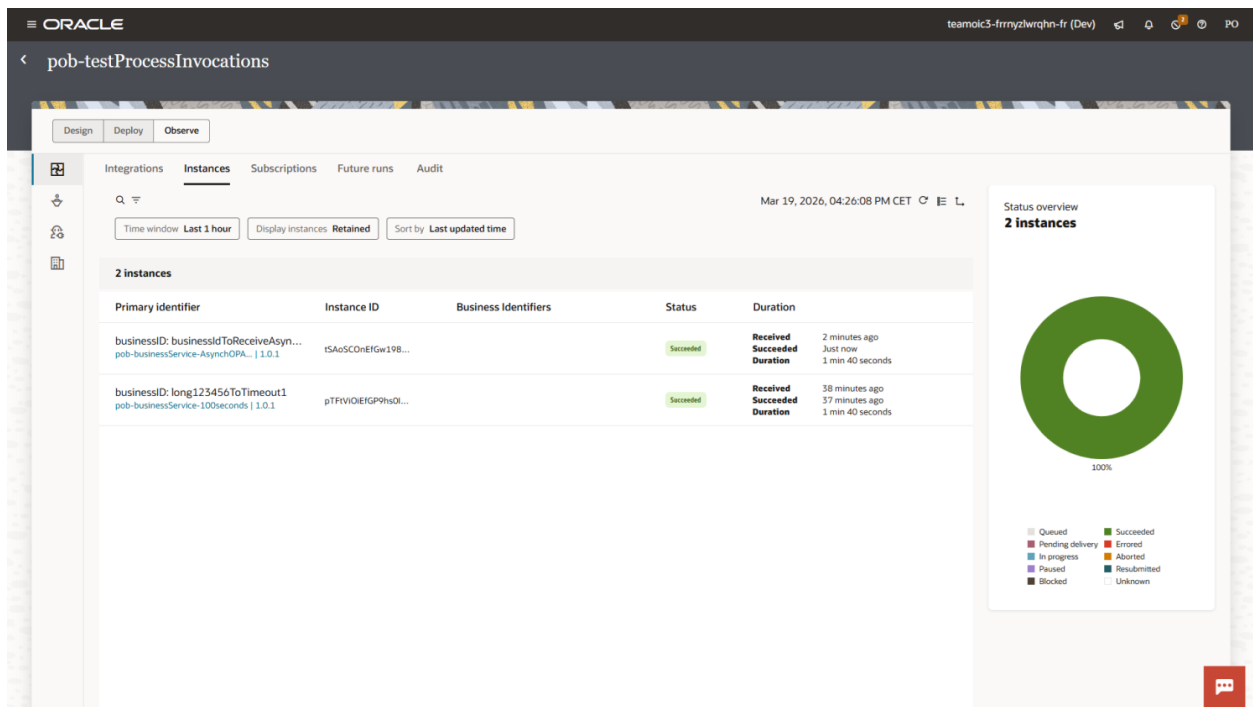
2 - Activate the new integration

Save, validate, and activate the cloned asynchronous integration. Keep the original synchronous version active until the new process is fully validated.

Part - 5 Testing and Validation

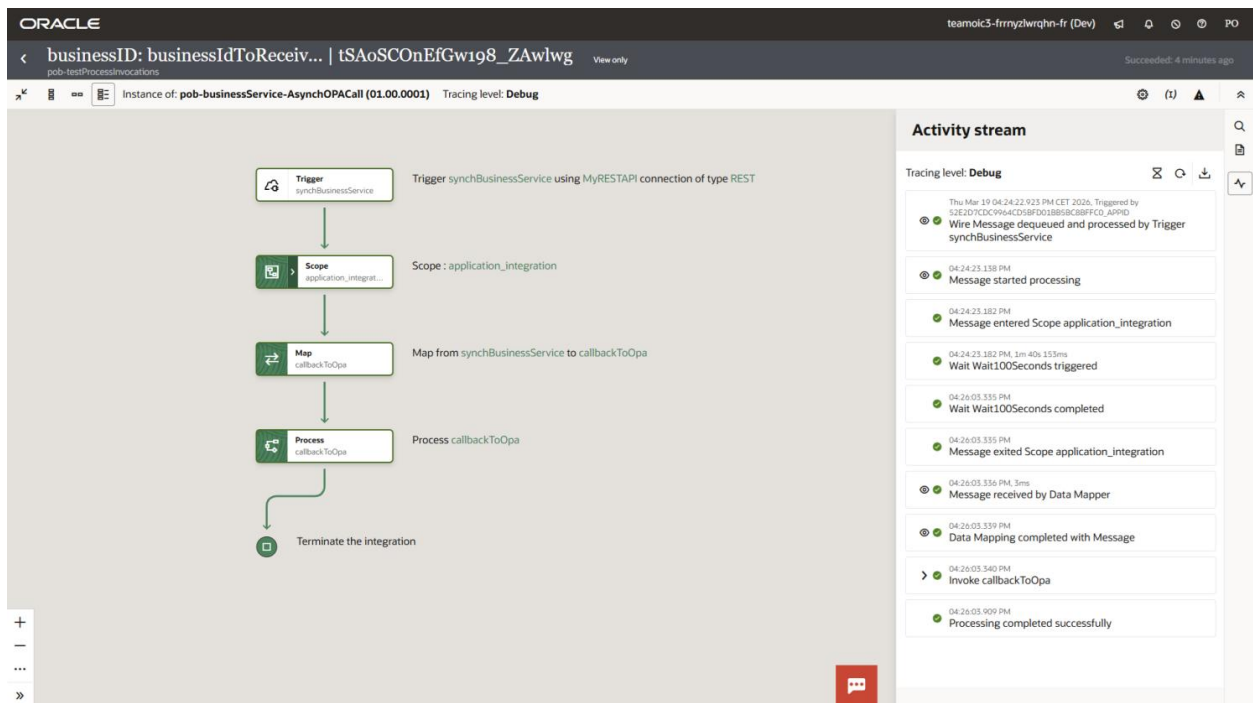
After activating all three components, run an end-to-end test to confirm the refactored flow works correctly.

- Initiate the refactored process from Process Workspace or your trigger mechanism.
- Confirm a new asynchronous Integration flow instance is initiated and completes successfully

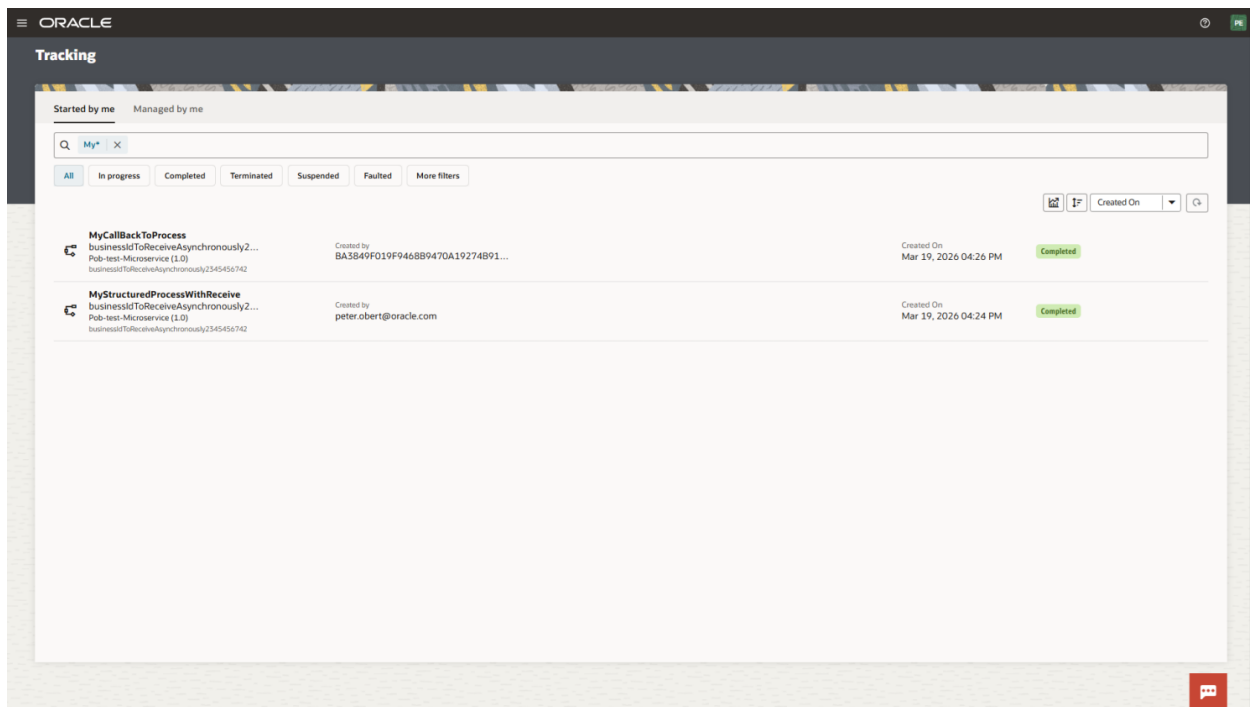


Example of the Synchronous and Asynchronous flow instances in the project

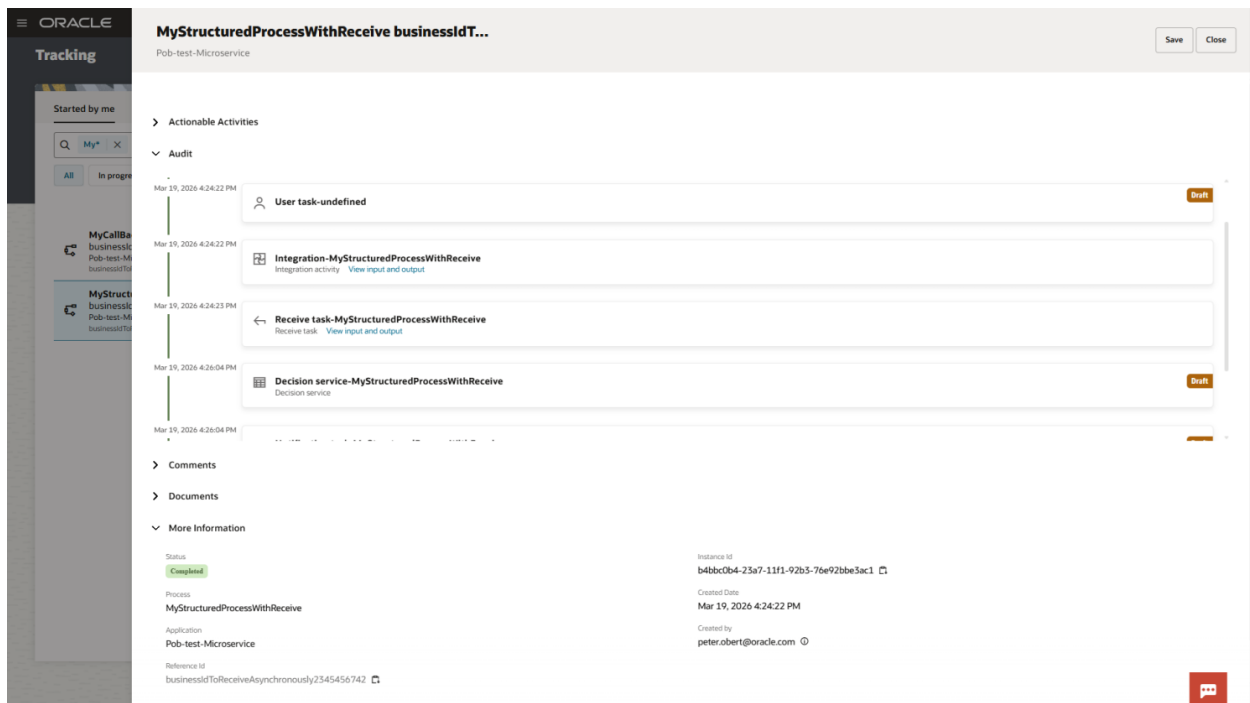
- Verify the Integration invokes the Callback Handler Flow on completion.



- Check Process Workspace: two process instances should show as Completed — the refactored business process and the Callback Handler.



- Confirm there are no timed-out instances in Process Workspace for the refactored process although business service call and task receive execution happens.



Notes

Why wrap business services in Integrations?

Isolating business service calls inside Integration orchestrations keeps your Process Automation flows technology-agnostic. Error handling, retries, and protocol-specific logic stay inside the integration layer, making the business process simpler and more maintainable.

Why not just raise the timeout limit?

Even if the OPA HTTP Connector timeout is increased beyond 90 seconds, synchronous design remains fragile for services that respond in minutes. An asynchronous pattern is inherently more resilient: the process instance persists and waits without holding a thread, and transient failures in the business service do not directly fault the process.

Correlation is critical

The Receive Task correlation properties are what route the callback payload back to the specific running process instance. Without correct correlation configuration, callbacks will either be lost or delivered to the wrong instance.

Templates

Integration project and Process application archive templates are soon available in the Oracle Technology Engineering GitHub Repository: github.com/oracle-devrel/technology-engineering

Credits: [Niall Commiskey](#), [Stan Tanev](#)